

Five-Wave Reliability

A local Screeps competition module

docs/competitions/02-five-wave-reliability.md

Summary

The learner runs repeated invader waves to test whether defense code is reliable, not just lucky once.

Skill Target

- Primary: regression testing under repeated combat pressure.
- Secondary: repair scheduling, tower energy logistics, creep replacement.
- Program track: Architect, Episode 13.
- Recommended episode: `docs/tutorials/13-sparring-ground.md`.

Prerequisites

- First Invasion Defense is passing.
- `scripts/sparring-loop.sh` is available.
- Room has tower refueling logic or enough stored energy.

Setup

Replace `W6N3` with the learner's room name.

```
scripts/sparring-loop.sh W6N3 5 10
```

This sends five waves with a ten-second delay between requests.

Challenge

Keep the colony alive through repeated waves without pausing to manually repair, refuel, or respawn.

Pressure

Five local sparring waves.

Win Condition

The learner passes when:

- `Spawn1` survives all waves.
- Tower energy never stays empty for more than one wave.
- Worker population recovers automatically after losses.
- The learner can name the weakest part of the defense loop.

Scoring

Metric	Full Credit	Partial Credit
Outcome	Survives all five waves and recovers.	Survives but stalls economy.
Code behavior	Defense, refuel, and repair priorities are explicit.	Defense works but logistics are fragile.
Debugging process	Learner compares wave 1 and wave 5 state.	Learner only reports final state.

Replay

Restore tower energy, then rerun the sparring loop. Keep the same delay when comparing attempts.

Reflection

- Did the same failure repeat?
- Which metric changed after your code fix?
- Was the bottleneck damage, energy, CPU, or spawning?

Extensions

- Easier: run three waves.
- Harder: reduce delay to five seconds.
- Team variant: one learner observes logs while another changes code.